Hyperspectral Anomaly Detection via Deep Plug-and-Play Denoising CNN Regularization

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***Abstract*— Due to the importance in many military and civilian applications, hyperspectral anomaly detection has attracted remarkable interest. Low-rank representation (LRR)-based anomaly detectors use the low-rank property to represent background pixels, and pixels that cannot be well represented are detected as anomalies. The ability of an LRR-based detector to separate background pixels and anomalous pixels depends on the dictionary representation ability, which usually can be enhanced by designing a proper prior for dictionary representation coefficients and constructing a better dictionary. However, it is not easy to handcraft effective and meaningful regularizers for dictionary coefficients. In this article, we propose a novel anomaly detection algorithm that uses a plug-and-play prior for representation coefficients and constructs a new dictionary based on clustering. Instead of cumbersomely handcrafting a regularizer for representation coefficients, we propose solving the anomaly detection problem using the plug-and-play framework, which enables us to plug state-of-the-art priors for representation coefficients. An effective convolutional neural network (CNN) denoiser is plugged into our framework to fully exploit the spatial correlation of representation coefficients. We also propose a modified background dictionary construction method, which carefully includes background pixels and excludes anomalous pixels from clustering results. We refer to the proposed anomaly detection method as plug-and-play denoising CNN regularized anomaly detection (DeCNN-AD) method. Extensive experiments were performed on five data sets in a comparison with eight state-of-the-art anomaly detection methods. The experimental results suggest that the proposed method is effective in anomaly detection and can produce better anomaly detection results than that of the comparison methods. The codes of this work will be available at https://github.com/FxyPd for the sake of reproducibility.**

***Index Terms*— Anomaly detection, convolutional neural network (CNN) denoiser, dictionary construction, hyperspectral image (HSI), plug-and-play.**

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I. Introduction

HYPERSPECTRAL cameras measure the electromagnetic energy scattered within their instantaneous field of view under hundreds or thousands of spectral channels, generating images with a remarkably high spectral resolution, which is the main difference with traditional visible or multispectral remote sensing images [1]. The high spectral resolution enables countless applications of hyperspectral images (HSIs), such as land cover classification [2], spectral unmixing [3], change detection [4], and target/anomaly detection [5], [6].

Due to the significance in many military and civilian applications, anomaly detection in HSIs has attracted a remarkable interest. Anomaly detectors aim to distinguish the potential targets from the background without prior knowledge of the spectral signal of the anomalies [7]. Since the spectral signal of the anomalies is unknown in the task of anomaly detection, it is very challenging to detect anomalies in HSI. Different from the background pixels, there are two discriminative characteristics of the anomalous pixels: they generally occur with a low probability compared with the background pixels, and their spectral signatures usually deviate from that of their surrounding background pixels [8]. These two characteristics of the anomalies allow the researchers to distinguish the anomalies from the backgrounds.

Over the past several decades, researchers have been devoted to improving the detection performance of the anomalies from the background in HSIs, and a lot of effective methods have been proposed. Among those methods, the most famous anomaly detection algorithm for HSIs is the Reed–Xiaoli (RX) method that is based on statistics [9]. The RX method detects the anomalies under the assumption that the background pixels follow a multivariate Gaussian distribution. The global RX (GRX) constructs the background using the whole pixels in the image, whereas the local RX (LRX) utilizes a local dual-window to simulate the background. In [10], the mean vector and covariance matrix were estimated under the Gaussian distribution to propose the weighted RX method. Considering that the background in the

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real-world is very complex, it is not enough to model the background using the Gaussian distribution. Many adaptations of the RX algorithm have been proposed to take into account this problem. Based on kernel theory, the kernel RX (KRX) algorithm [11] projects the data into a high-dimensional feature space so that it is able to characterize the background under non-Gaussian distributions. By dividing the image into different clusters, the cluster-based anomaly detection (CBAD) algorithm detects the anomalies within each cluster [12]. When anomalies and background are hard to distinguish in original HSIs, images can be projected into other spaces to enlarge the difference between the anomalies and the background. Statistics-based detection methods are then applied to detect the anomalies, such as probabilistic anomaly detector [13], anomaly detector based on subspace projection [14], discriminative metric-learning-based detector [15], and fractional Fourier (FrFE)-based method [16].

In addition to statistics-based methods, various kinds of anomaly detection methods have been proposed in the literature. Tu *et al.* [17] proposed an anomaly detector, which is based on the dual window density (DWD) and density peak clustering (DPC) algorithm. Kang *et al.* [18] proposed a novel detection method, which utilized the attribute and edge-preserving filter to detect anomalies, and its multiscale version has been developed in [19]. A novel kernel isolation forest-based hyperspectral anomaly detection (KIFD) method was proposed in [20], and a fast anomaly detector for hyperspectral imagery based on multiple features and isolation forest was proposed in [21]. Lei *et al.* [22] proposed a hyperspectral anomaly detection method based on spectral learning and discriminative reconstruction (SLDR). Meanwhile, subspace selection-based discriminative forest (SSDF) method [23], locally enhanced low-rank prior-based anomaly detector (LELRP-AD) [24], nonlinear spectral-spatial composite kernel-based detector (SSCKD) [25], prior-based tensor approximation (PTA)-based method [26], and image segmentation-based method [27] have been proposed to detect anomalous pixels in HSIs. In addition, deep-learning-based methods have attracted great attention [28]. The unsupervised learning method [29], semisupervised learning [30], and transferred learning method [31], [32] have been explored to detect anomalies in HSIs. Generally, the deep-learning-based methods need large a large amount of label data and a long time to train the networks. Once the networks are trained, they can usually achieve desirable detection performance in a short time. Compared with deep-learning-based methods, machine learning-based detectors do not need any training data, but they may take some time when applied to various HSIs and their parameters need to be tuned manually.

Recently, representation-based methods have attracted considerable interest for hyperspectral anomaly detection. Based on the assumption that background pixels can be approximately represented by pixels surrounded within a predefined dual window, whereas the anomalous pixels cannot, a collaborative representation-based detector (CRD) has been proposed in [33]. To better exploit the spatial information, Imani [34] proposed the CRD based on the morphology (MCRD). The principal component analysis with CRD (PCAroCRD) to detect anomalies was proposed to improve the judgment in [35]. To facilitate the computational efficiency of the CRD algorithm, the recursive CRD (RCRD) method was proposed [36], in which the update was derived recursively by exploiting a matrix inversion lemma. By incorporating the learning of weight and the collaborative representation into consideration, Wang *et al.* [37] proposed a self-weighted collaborative representation-based detector (SWCRD).

Instead of detecting anomaly pixel-by-pixel, RPCA [38] decomposes the HSIs into low-rank and sparse parts by making use of the high correlations along spectral dimension. Different from RPCA, the low-rank representation (LRR) model [39] deals with data containing outliers in multiple subspaces, which makes it more appropriate to recover more complex scene. The LRR model was first introduced to handle anomaly detection in HSIs in low rank and sparse representation (LRASR) method [7], in which an ℓ_1 sparse norm was introduced to constrain the representation coefficients of the dictionary. However, these two regularizers used concurrently in LRASR method may be redundant to some degree [40]. In [8], a new anomaly detection model based on LRR and collaborative representation via robust dictionary construction (LRCRD) was proposed, in which representation coefficient term was regularized by a nuclear norm and a weighted ℓ_2-norm to combine global structure and local attributes of images. To further take into account the potential spatial information, which depicts the spatial position between the test pixel and its neighbors, Cheng and Wang [40] proposed a graph and total variation (TV) regularized LRR (GTVLRR) method for hyperspectral anomaly detection. In this work, the graph Laplacian was derived based on the graph theory and was exploited to regularize the representation coefficient and preserve the local manifold structure with the increase in computation time.

Considering the inherent spectral and spatial information in the representation coefficients, it is important to define powerful priors for the representation coefficients to enhance the performance of LRR-based anomaly detectors. However, it is not easy to handcraft effective and meaningful regularizers, and complex regularizers may increase the difficulty of solving the optimization problems. Fortunately, benefitting from the variable splitting techniques, plug-and-play priors have been exploited to solve various inverse problems. The main idea of using plug-and-play priors is to use state-of-the-art priors from off-the-shelf denoisers (such as BM3D [41], WNNM [42], and convolutional neural network (CNN)-based denoisers [43], [44]), instead of creating new priors. The superiority of plugged priors has been demonstrated in solving hyperspectral inverse problems, e.g., compressive imaging [45], image denoising [46], [47], and image fusion [48].

In this article, we propose a novel anomaly detection algorithm based on the plug-and-play framework. To avoid cumbersomely handcrafting a regularizer that may not be effective for regularizing the representation coefficient or may be complex to optimize, we propose a new detector plugged with a prior for representation coefficients. The plugged prior is extracted from a CNN. Furthermore, to obtain a background dictionary without anomaly contamination, we propose

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a modified dictionary construction method based on clustering. We refer to the proposed anomaly detection method as plug-and-play denoising CNN regularized anomaly detection (DeCNN-AD) method. The main contributions of our work can be summarized as follows.

1) To the best of our knowledge, this is the first work using plugged priors for hyperspectral anomaly detection. Instead of trivially exerting a handcrafted regularizer on the representation coefficients, we propose solving the anomaly detection problem using the plug-and-play framework, which enables us to plug state-of-the-art priors for representation coefficients.
2) A state-of-the-art CNN-based denoiser, fast and flexible denoising convolutional neural network (FFDNet), is used as the prior in the plug-and-play framework. The plugged CNN prior is proved to be effective in regularizing the representation coefficients without retraining.
3) We propose a modified background dictionary construction method, which carefully includes background pixels and excludes anomalous pixels. Experiments show that a more pure and comprehensive dictionary can boost the performance of anomaly detection.

Extensive experiments were performed on five data sets in a comparison with eight state-of-the-art anomaly detection methods to thoroughly evaluate the effectiveness of the proposed method.

The remainder of this article is organized as follows. Section II-A briefly reviews the related work of anomaly detection. Section III introduces our new anomaly detection algorithm in detail. Section IV presents and analyzes the experimental results, including the comparisons with the state-of-the-art methods. Section V concludes this article.

II. Problem Formulation and Related Work

A. Observation Model

Let $\mathbf{X} \in \mathbb{R}^{L\times N}$ denote an observed HSI, where L represents the number of bands and N denotes the number of pixels in HSI. As anomalous pixels are spectrally different from the background pixels, the observed image can be decomposed into the background component, $\mathbf{B} \in \mathbb{R}^{L\times N}$, and the anomalous component, $\mathbf{E} = [\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_N] \in \mathbb{R}^{L\times N}$, that is

$$\mathbf{X} = \mathbf{B} + \mathbf{E}. \tag{1}$$

Because of the high correlation among background pixels, the background component can be sparsely represented in a dictionary. We have

$$\mathbf{B} = \mathbf{DS} \tag{2}$$

where $\mathbf{D} = [\mathbf{d}_1, \mathbf{d}_2, \ldots, \mathbf{d}_m] \in \mathbb{R}^{L\times m}$ represents a dictionary, whose columns are spectral vectors of m pixels selected from the background pixels. Matrix $\mathbf{S} = [\mathbf{s}_1, \mathbf{s}_2, \ldots, \mathbf{s}_N] \in \mathbb{R}^{m\times N}$ holds representation coefficients of $\mathbf{B}$ with respect to (w.r.t.) $\mathbf{D}$. The observation model is illustrated in Fig. 1.

B. Related Work

1) LRR: The LRR model aims to find the LRR of the background component $\mathbf{B}$ w.r.t. the dictionary $\mathbf{D}$. By choosing a convex relaxation of the rank of representation coefficients $\mathbf{S}$, the LRR model can be written as

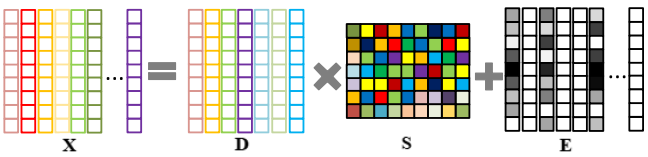


Fig. 1. Observation model of the LRR-based anomaly detection model.

$$\{\widehat{\mathbf{S}}, \widehat{\mathbf{E}}\} \in \min_{\mathbf{S},\mathbf{E}} \|\mathbf{S}\|_* + \lambda\|\mathbf{E}\|_{2,1}, \quad \text{s.t. } \mathbf{X} = \mathbf{DS} + \mathbf{E} \tag{3}$$

where $\|\cdot\|_*$ denotes the matrix nuclear norm, $\|\cdot\|_{2,1}$ denotes the $\ell_{2,1}$-norm, and λ is a parameter imposed on the $\|\cdot\|_{2,1}$-norm. If a proper dictionary is chosen, LRR can separate the background and detect anomalies as well. Once matrices $\mathbf{D}$ and $\mathbf{S}$ are estimated by solving the problem (3), the anomalous degree of the ith spectrum, denoted as $T(\mathbf{x}_i)$, can be determined by calculating the signal power of the columns in $\mathbf{E}$ as follows:

$$T(\mathbf{x}_i) = \|\hat{\mathbf{e}}_i\|_2 \tag{4}$$

where $\hat{\mathbf{e}}_i$ denotes the ith column of matrix $\widehat{\mathbf{E}}$. If $T(\mathbf{x}_i)$ is larger than a given threshold, then pixel $\mathbf{x}_i$ is considered to be an anomaly.

2) LRASR: The low rankness constrain is good at characterizing the global structure but not the local structure. To make better use of the local structure information, a sparse norm is introduced in LRASR [7]. Global structure is regularized by the low-rank term, and the local structure is exploited by the sparsity term. LRASR solves the following optimization problem:

$$\min_{\mathbf{S},\mathbf{E}} \|\mathbf{S}\|_* + \beta\|\mathbf{S}\|_1 + \lambda\|\mathbf{E}\|_{2,1} \quad \text{s.t. } \mathbf{X} = \mathbf{DS} + \mathbf{E} \tag{5}$$

where $\|\cdot\|_1$ denotes the ℓ_1-norm of a matrix; β and λ are parameters to tradeoff low rankness and sparsity. To avoid expensive matrix inversions during the optimization, LRASR adopts the linearized alternating direction method with adaptive penalty (LADMAP) method to solve (5).

3) LRCRD: Based on the theory of collaborative representation, it is considered that the collaboration between dictionary atoms is more important than sparsity. Thus, the LRCRD algorithm [8] exerts a weighted ℓ_2-norm on the representation coefficient to promote as many dictionary atoms as possible to participate in the representation, which is different from the LRSAR method. The LRCRD solved the following optimization problem:

$$\min_{\mathbf{S},\mathbf{E}} \|\mathbf{S}\|_* + \beta\|\mathbf{W} \circ \mathbf{S}\|_2 + \lambda\|\mathbf{E}\|_{2,1} \quad \text{s.t. } \mathbf{X} = \mathbf{DS} + \mathbf{E} \tag{6}$$

where $\mathbf{W} \in \mathbb{R}^{m\times N}$ represents a weight matrix and $w_{j,i}$ denotes the Euclidean distance between the jth atom of D and the ith pixel of $\mathbf{X}$. The $\circ$ denotes the Hadamard product between $\mathbf{W}$ and $\mathbf{S}$.

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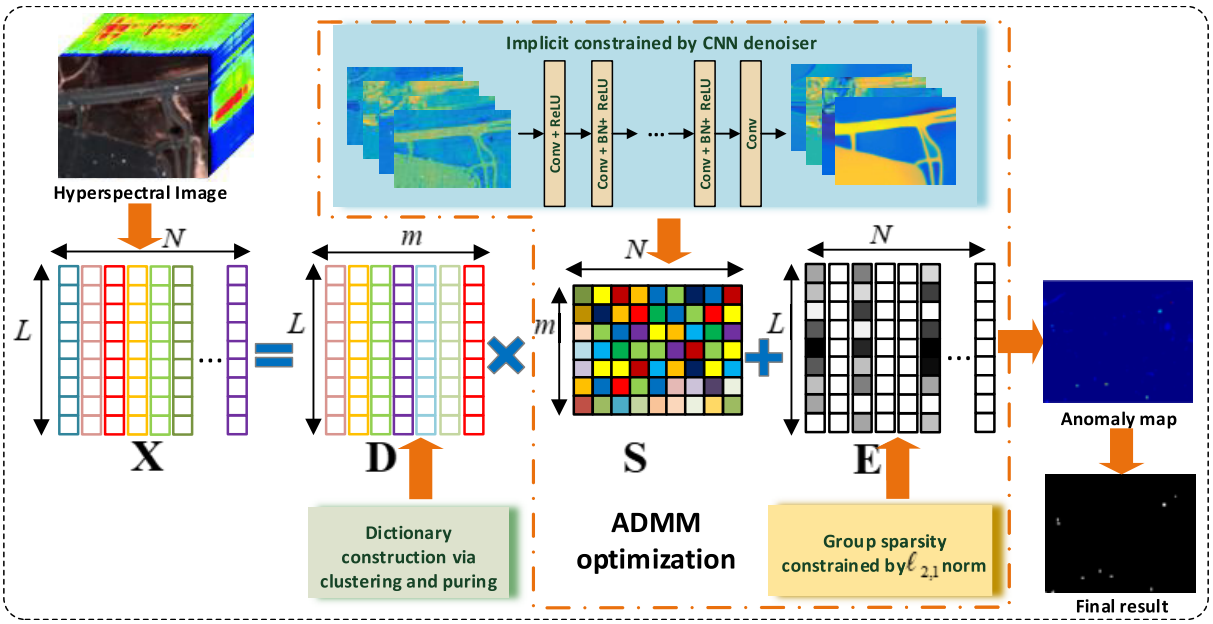


Fig. 2. Illustration of the proposed anomaly detection method.

4) GTVLRR: Both the LRASR and the LRCRD utilize spectral signature to detect the anomalies and ignore the spatial information. To further incorporate the spatial information into the anomaly detection, graph and TV regularizations were introduced in the GTVLRR method [40], which minimizes the following optimization:

$$\min_{\mathbf{S},\mathbf{E}} \|\mathbf{S}\|_* + \beta\|\mathbf{HS}\|_1 + \gamma\,\mathrm{Tr}(\mathbf{SLS}^{\mathbf{T}}) + \lambda\|\mathbf{E}\|_{2,1} \quad \text{s.t. } \mathbf{X} = \mathbf{DS} + \mathbf{E} \tag{7}$$

where β, λ, and γ are parameters to tradeoff the different regulation terms. $\mathrm{Tr}(\mathbf{SLS}^{\mathbf{T}})$ can be expressed as

$$\mathrm{Tr}\left(\mathbf{SLS}^{\mathbf{T}}\right) = \frac{1}{2}\sum_{i,j=1}^{N} \left\|\mathbf{s}_i - \mathbf{s}_j\right\|_2^2 W_{ij} \tag{8}$$

where $\mathbf{L} = \mathbf{M} - \widehat{\mathbf{W}}$ is the graph Laplacian matrix, $\widehat{\mathbf{W}}$ is the weight matrix of the graph G, $\mathbf{M}$ is a diagonal matrix with entries measuring the degree of each pixel, and $M_{ii} = \sum_{j=1}^{N} \widehat{W}_{ij}$. If a pixel $\mathbf{x}_i$ is among the k-nearest neighbors of pixel $\mathbf{x}_j$ or pixel $\mathbf{x}_j$ is among the k-nearest neighbors of pixel $\mathbf{x}_i$, then

$$\widehat{W}_{ij} = \exp\left(-\frac{\left\|\mathbf{x}_i - \mathbf{x}_j\right\|_2^2}{\sigma}\right) \tag{9}$$

where σ is the scalar parameter. $\|\mathbf{HX}\|_{1,1}$ denotes the TV term and can be expressed as

$$\|\mathbf{HX}\|_{1,1} = \left\|\begin{bmatrix}\mathbf{H_h X} \\ \mathbf{H_v X}\end{bmatrix}\right\|_{1,1} \tag{10}$$

where $\mathbf{H_h}$ denotes the linear operator calculating the differences of representation coefficients between each pixel and its horizontal neighboring pixel, and $\mathbf{H_v}$ denotes the difference between each pixel and its vertical neighboring pixel.

III. Proposed DeCNN-AD Method

From the related work in the previous section, it can be seen that the LRR-based anomaly detection methods achieve different anomaly detection performance by defining different priors for the representation coefficients. However, it is not trivial to define a prior that can completely incorporate the characteristics embedded in the representation coefficients. What is more, the optimization of the problem may be difficult if a complex prior is introduced. Thus, in this article, instead of manually handcrafting a prior, we introduce the plug-and-play framework, which allows us to use a denoiser as the prior of the representation coefficients. Meanwhile, a modified dictionary construction method is proposed to obtain a more pure background dictionary excluding anomalies. The illustration of the proposed anomaly detection is presented in Fig. 2.

A. Plug-and-Play Denoising CNN Constrained Anomaly Detection Model

The objective function of the proposed method is written as follows:

$$\min_{\mathbf{S},\mathbf{E}} \frac{1}{2}\|\mathbf{X} - \mathbf{DS} - \mathbf{E}\|_F^2 + \beta\phi(\mathbf{S}) + \lambda\|\mathbf{E}\|_{2,1} \tag{11}$$

where $\phi(\cdot)$ is the regularizer defined to further promote the desired property of representation coefficients $\mathbf{S}$, and β and λ are parameters to trade off the regulation of $\mathbf{S}$ and the group sparsity property of $\mathbf{E}$. Note that here we do not give an explicit definition of the function $\phi(\cdot)$ because we use plug-and-play prior [49] for $\mathbf{S}$. The central idea in plug-and-play is to use directly an off-the-shelf denoiser conceived for image denoising, instead of investing efforts in tailoring regularizers for images and then computing its proximity operators. As deep learning technique has been applied to image denoising area and shows promising results, we plug

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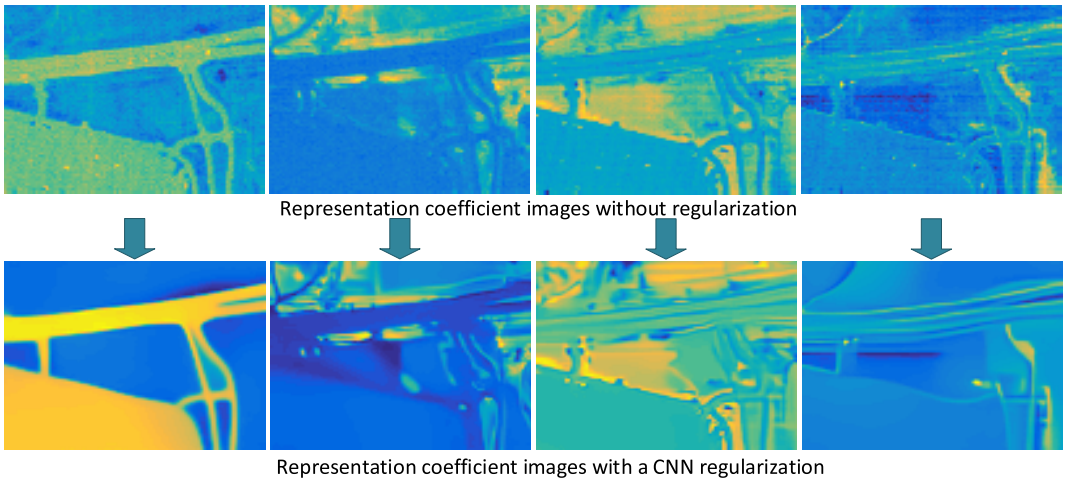


Fig. 3. Representation coefficient images of the HYDICE Urban data set without and with a regularization.

a CNN prior in (11) to exploit the high spatial correlation in $\mathbf{S}$ (see Section III-A3 for further details).

The high spatial correlation of dictionary coefficients motivates us to use a CNN prior for $\mathbf{S}$. We take a real HSI, HYDICE Urban image, as an example. If we do not impose any regularization on dictionary coefficients, we can obtain a coarse estimate of $\mathbf{S}$, whose rows are coefficients corresponding to atoms in the dictionary. We randomly selected four row vectors from the coarse estimate of $\mathbf{S}$ and reshaped them as 2-D images, as shown in the first row of Fig. 3. We can see that the coefficient images are spatially correlated and contain random noise and anomaly information, which exists because of the imperfection of the constructed background dictionary and may be removed by the plugged denoiser. The second row of Fig. 3 shows coefficient images obtained by our proposed method plugged with the FFDNet denoiser. We can see that the plugged denoiser reduces the noise and background representation errors in the coefficient images, producing smoother and cleaner dictionary coefficient images. Thus, background pixels can be better represented using the denoised coefficients. The better the background pixels are represented, the better the separability of the background and the anomalies we can get. As a result, we can achieve better detection performance.

Since the objective function (11) is not separable, by introducing an auxiliary variable $\mathbf{Z}$, (11) can be converted to into a constrained formulation

$$\min_{\mathbf{S},\mathbf{E},\mathbf{Z}} \frac{1}{2}\|\mathbf{X}-\mathbf{DS}-\mathbf{E}\|_F^2 + +\beta\phi(\mathbf{Z}) + \lambda\|\mathbf{E}\|_{2,1} \quad \text{s.t. } \mathbf{Z}=\mathbf{S}. \tag{12}$$

The augmented Lagrangian function of problem (12) can be expressed as follows:

$$L(\mathbf{S},\mathbf{E},\mathbf{Z},\mathbf{V}) = \frac{1}{2}\|\mathbf{X}-\mathbf{DS}-\mathbf{E}\|_F^2 + \beta\phi(\mathbf{Z}) + \lambda\|\mathbf{E}\|_{2,1} + \frac{\mu}{2}\left\|\mathbf{Z}-\mathbf{S}+\frac{\mathbf{V}}{\mu}\right\|_F^2 \tag{13}$$

where μ is the penalty parameter and $\mathbf{V}$ is the Lagrangian Multiplier. The optimization of (13) can be solved effectively by using the alternative method of multipliers (ADMM), which splits (13) into several subproblems and updates each variable iteratively.

1) Updating of $\mathbf{S}$*:* During the iteration, $\mathbf{S}$ is updated by minimizing $L(\mathbf{S},\mathbf{E},\mathbf{Z},\mathbf{V})$ w.r.t $\mathbf{S}$, which can be expressed as

$$\mathbf{S} \in \arg\min_{\mathbf{S}} L(\mathbf{S},\mathbf{E},\mathbf{Z},\mathbf{V}) = \arg\min_{\mathbf{S}} \frac{1}{2}\|\mathbf{X}-\mathbf{DS}-\mathbf{E}\|_F^2 + \frac{\mu}{2}\left\|\mathbf{Z}-\mathbf{S}+\frac{\mathbf{V}}{\mu}\right\|_F^2 \tag{14}$$

which is strongly convex. Thus, $\mathbf{S}$ can be obtained by calculating the derivative of (14) with respect to $\mathbf{S}$ and set it to be zero

$$\mathbf{S} = (\mathbf{D}^T\mathbf{D}+\mu\mathbf{I})^{-1}(\mathbf{D}^T(\mathbf{X}-\mathbf{E})+\mu\mathbf{Z}+\mathbf{V}). \tag{15}$$

2) Updating of $\mathbf{E}$*:* During the iteration, $\mathbf{E}$ is updated by minimizing $L(\mathbf{S},\mathbf{E},\mathbf{Z},\mathbf{V})$ w.r.t $\mathbf{E}$ and can be expressed as

$$\mathbf{E} \in \arg\min_{\mathbf{E}} L(\mathbf{S},\mathbf{E},\mathbf{Z},\mathbf{V}) = \arg\min_{\mathbf{E}} \frac{1}{2}\|\mathbf{X}-\mathbf{DS}-\mathbf{E}\|_F^2 + \lambda\|\mathbf{E}\|_{2,1} \tag{16}$$

which has a closed-form solution by using the $\ell_{2,1}$-norm minimization operator [39]

$$\mathbf{Z} = \text{shrink}_{2,1}(\mathbf{X}-\mathbf{DS},\lambda) \tag{17}$$

where

$$\text{shrink}_{2,1}(\mathbf{U},\lambda) = [\text{shrink}(\mathbf{u}_i,\lambda), i=1,\ldots,N]. \tag{18}$$

The function $\text{shrink}(\mathbf{u},\lambda)$ is a vector-soft-threshold operator defined as

$$\mathbf{u} \mapsto \frac{\max(\|\mathbf{u}\|-\lambda,\mathbf{0})}{\max(\|\mathbf{u}\|-\lambda,\mathbf{0})+\lambda}\mathbf{u}.$$

3) Updating of $\mathbf{Z}$*:* During the iteration, $\mathbf{Z}$ is updated by minimizing $L(\mathbf{S},\mathbf{E},\mathbf{Z},\mathbf{V})$ w.r.t $\mathbf{Z}$, which can be expressed as

$$\mathbf{Z} \in \arg\min_{\mathbf{Z}} L(\mathbf{S},\mathbf{E},\mathbf{Z},\mathbf{V}) = \arg\min_{\mathbf{Z}} \beta\phi(\mathbf{Z}) + \frac{\mu}{2}\left\|\mathbf{Z}-\mathbf{S}+\frac{\mathbf{V}}{\mu}\right\|_F^2. \tag{19}$$

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Based on the spirit of the plug-and-play, we can find that the optimization of (19) is a denoising operator of ϕ, in which the input noisy data are $(\mathbf{S}-\mathbf{V}/\mu)$, the input noise level is $\sigma^2 = \beta/\mu$, the prior regularization term is $\phi(\mathbf{Z})$, and the data fidelity term is $(\mu/2)\|\mathbf{Z}-\mathbf{S}+(\mathbf{V}/\mu)\|_F^2$. As a result, the solution of the minimization of problem (19) can be denoted as

$$\mathbf{Z} \leftarrow \text{Denoiser}\left(\mathbf{S}-\frac{\mathbf{V}}{\mu}, \frac{\beta}{\mu}\right) \tag{20}$$

where Denoiser$(\cdot)$ is a plugged state-of-the-art denoiser, which can be any off-the-shelf denoiser (such as BM3D [41], BM4D [50], NLM [51], WNNM [42], and CNN-based denoising methods [43], [44]), instead of investing great effort in designing new regularizers for image denoising. The replacement of (19) by (20) is called the plug-and-play step [46], [49].

In particular, the rapid development of deep learning has raised new opportunities to deal with different image processing problems, which were generally handled by using traditional mathematical or physical models before. Due to the powerful nonlinear learning and representing capabilities of image features, CNNs have achieved great success in various kinds of image restoration problems. The CNNs learn various spatial-spectral features of images by using lots of neurons in the networks to imitate the learning process of the human brain. Compared with the prior terms (such as sparsity and low-rank regularizations) used in traditional machine learning (ML)-based denoisers to characterize the inherent information in images, the prior information learned by CNNs has an implicit form without the need for giving an explicit mathematical expression to characterize the prior features. If the data set used to train the CNN is large enough, the trained CNN can learn deep and nonlinear features for images, which may boost the performance of traditional ML-based denoisers. What is more, the well trained CNN can also run faster than many traditional denoising methods, such as BM3D and WNNM.

Due to powerful learning capabilities of CNN, a fast and stable CNN denoised method, FFDNet [43], is used as the plugged denoiser in (20). There are a series of 3×3 convolution layers in the FFDNet. Each layer is composed of three types of operations: convolution (Conv), rectified linear units (ReLU), and batch normalization (BN) [43]. Specifically, the first layer is "Conv + ReLU," the middle layers are "Conv + BN + ReLU," and the last layer is "Conv," where BN is exploited to speed up the training procedure, and ReLU is the activation function. The FFDNet decomposes the noisy image as four subimages and uses these subimages as the input to reduce the number of layers and speed the denoising up. By taking a tunable noise level map as input, the FFDNet is able to deal with noise of different levels. The advantages of using FFDNet are threefold. First, the FFDNet has been trained on a large number of available images and, thus, can provide stable denoising results. Second, FFDNet can address images with various noise levels, meaning that we can input a new image without retraining. Third, FFDNet works on downsampled subimages, which largely accelerates the denoising speed while maintaining the performance.. As $\mathbf{S}$ is the representation coefficients of the HSI, each row of $\mathbf{S}$ preserves spatial information of the HSI, as shown in Fig. 3. Thus, the well-trained FFDNet is used to denoise each row of $(\mathbf{S}-\mathbf{V}/\mu)$ to solve the problem, i.e., we solve the problem (19) by applying the denoising method on each row of $(\mathbf{S}-\mathbf{V}/\mu)$. More specifically, $\mathbf{Z}$ is solved row by row by using a map function $\boldsymbol{\Omega}$ learned from FFDNet and can be expressed as

$$\mathbf{Z}(i,:) = \boldsymbol{\Omega}(\mathbf{H}(i,:), \frac{\beta}{\mu}; \Theta), \quad i = 1, 2, \ldots, m \tag{21}$$

where Θ is the learned parameter of the FFDNet, β/μ is the input noise level, and $\mathbf{H} = \mathbf{S} - \mathbf{V}/\mu$. We first normalized each row of $\mathbf{H}$ to the range of [0, 1] and then reshaped it to be a matrix denoted as $\hat{\mathbf{H}}_i$ with the size of $W \times H$, which are the row and column of the HSI, respectively. The operation on each row of $\mathbf{H}$ can be represented as $\hat{\mathbf{H}}_i = \text{vec}^{-1}(c_i\mathbf{H}(i,:) + b_i)$, where $c_i = (1/(\max(\mathbf{H}(i,:)) - \min(\mathbf{H}(i,:))))$, $b_i = -c_i * \min(\mathbf{H}(i,:))$, and $\text{vec}^{-1}(\cdot)$ denotes the inverse operation of the vectorization, which reshapes a vector to a matrix. For example, for a matrix $\mathbf{G} \in \mathbb{R}^{W\times H}$ and a vector $\mathbf{a} \in \mathbb{R}^{WH}$, $\mathbf{G} = \text{vec}^{-1}(\mathbf{a})$ is the same as $\mathbf{G}(i,j) = \mathbf{a}((j-1)*W + i)$. As we normalized the data, the noise level should also be changed to $c_i\sigma^2$ accordingly. After the denoising, we need to reshape and rescale the denoised result back. To summarize, we solve the optimization of $\mathbf{Z}$ as follows:

$$\hat{\mathbf{H}}_i = \text{vec}^{-1}(c_i\mathbf{H}(i,:) + b_i), \quad i = 1, 2, \ldots, L \tag{22}$$

$$\hat{\mathbf{Z}}_i = \text{vec}\left(\boldsymbol{\Omega}\left(\hat{\mathbf{H}}_i, \frac{\beta}{\mu}; \Theta\right), 1, N\right), \quad i = 1, 2, \ldots, L \tag{23}$$

$$\mathbf{Z}(i,:) = \frac{\hat{\mathbf{Z}}_i - b_i}{c_i}, \quad i = 1, 2, \ldots, L. \tag{24}$$

4) Updating of the Lagrangian Multiplier $\mathbf{V}$*:* During the iteration, the Lagrangian multiplier $\mathbf{V}$ is updated as follows:

$$\mathbf{V} = \mathbf{V} + \mu(\mathbf{Z} - \mathbf{S}). \tag{25}$$

5) Updating of the Penalty Parameter μ*:* As can be seen that at the denoising step in each iteration, the input noise variance of the FFDNet is decided by μ and β, i.e., $\sigma^2 = \beta/\mu$, which means that the penalty parameter μ plays important role in the denoising process. As the iteration continues, the image is denoised iteratively, and the noise left in the image becomes lesser. Thus, the noise level input into the FFDNet should be smaller after each iteration to adapt to the noise left in the image. The input noise level can be tuned down by increasing the value of μ after each iteration. The penalty is updated by

$$\mu = \min(\rho\mu, \mu_{\max}) \tag{26}$$

where ρ is a parameter larger than 1. In this article, ρ is set to 1.1.

The algorithm for solving the objective function (11) is summarized in Algorithm 1.

B. Construction of Dictionary $\mathbf{D}$

For the anomaly detection task, the dictionary and the anomalous targets are not known in advance, so construction of dictionary is of importance and should be very careful.

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Algorithm 1 Proposed DeCNN-AD Model

Input: $\mathbf{X}$, $\mathbf{D}$, β, λ, ρ
While not converge **do**
  Update $\mathbf{S}$ via (15)
  Update $\mathbf{E}$ via (17)
  Update $\mathbf{Z}$ via (22) $\sim$ (24)
  Update $\mathbf{V}$ via (25)
  Update μ via (26)
end while
Output S, **E**

In order to split the image into the background component and anomalous component, the background dictionary should be able to represent all background pixels and exclude the anomalous component. Based on the fact that anomaly pixels have spectral signals significantly different from the backgrounds and they normally occur with a low probability, we proposed a modified dictionary construction method selecting background pixels from image clustering results.

The procedure of background dictionary construction is summarized in Algorithm 2. Pixels from the original image are divided into K groups using the well-known K-means clustering method [52]. To remove suspicious anomaly pixels within each cluster, we design an inner iteration. For the jth iteration, we detect and remove $M \times n_i^l$ pixels with the largest Mahalanobis distance to the current center vector of the ith cluster. As being far away from the center of the cluster, the detected pixels are at high risk of belonging to the anomalous component, but not the background component. The removal of suspicious pixels in the ith cluster repeats T times. After that, the anomaly pixels can be considered to be removed completely. Finally, for each cluster with remaining pixels more than P, the pixels with P smallest Mahalanobis distance to the current center vector are chosen and added into the dictionary. By doing this, we could help improve the representation capability of the dictionary for the background pixels. In this article, parameters, T, M, and P, are empirically fixed to 3, 0.3, and 20, respectively.

The dictionary construction method proposed in Algorithm 2 is inspired by LRASR [7], which also selects background pixels from clustering results. A challenge of clustering-based dictionary construction is to exclude anomalous component. If an anomaly pixel is wrongly added into the dictionary, then the anomaly will be detected with a higher false alarm rate (FAR). This motivates us to design a new dictionary construction, which excludes anomalous pixels by designing an inner iteration. An experimental comparison between two dictionary construction methods is given in Section IV-F.

IV. Experiments

Extensive experiments were carried out to assess the performance of the proposed DeCNN-AD method for HSI anomaly detection in comparison with eight state-of-the-art HSI anomaly detectors. The results were evaluated with respect to both the visualization and numerical evaluating indexes.

Algorithm 2 Background Dictionary Construction

Input: data matrix $\mathbf{X}$, the number of clusters K
Initialize: $\mathbf{D} = \varnothing$, $T = 3$, $M = 0.3$, $P = 20$
$\{\mathbf{X}^1, \mathbf{X}^2, \ldots, \mathbf{X}^K\} = \text{kmeans}(\mathbf{X}, K)$, where the columns of $\mathbf{X}^i \in \mathbb{R}^{L \times n_i}$ contain n_i pixels belonging to ith cluster.
for $i = 1: K$ **do**
  for $l = 1: T$ **do**
    Compute the mean vector μ and covariance matrix of cluster $\mathbf{X}^i$
    Remove $M \times n_i^l$ pixels with largest Mahalanobis distance to the center vector of the ith cluster.
  end for
  if $n_i^T < P$ **then**
    Skip to Step 4
  end if
  Compute Mahalanobis distance to the center vector for every pixels.
  Add pixels with P smallest Mahalanobis distance to the center vector into the dictionary $\mathbf{D}$.
end for
Output D

A. Hyperspectral Data Sets

The performance of the proposed DeCNN-AD method is assessed on five real hyperspectral data sets that were captured at different scenes. The detail of the data sets is given as follows.

1) The first data set is the Pavia Center data set that was captured in the Pavia city center in Italy. The data set was collected by a reflective optics system imaging spectrometer (ROSIS) sensor with a spatial resolution of 1.3 m. The spectral signal range from 430 to 860 nm and was divided into 102 spectral bands. The scene used in the experiment is a scene with 80 $\times$ 100 pixels. The vehicles on the bridge are the anomalies to be detected from the bridge and water. We show the false-color image of the data set and its ground-truth map of anomalous targets in Fig. 4(a).
2) The second data set is Hyperspectral Digital Imagery Collection Experiment (HYDICE) data set. It is captured by the HYDICE airborne sensor with a spatial resolution of 1 m over an urban area in the USA. The scene used has a spatial size of 80 $\times$ 100 pixels with 175 spectral bands, whose wavelengths ranging from 400 to 2500 nm. The scene consists of a vegetation area, a construction area, and several roads with a few vehicles on them. The anomalies are man-made objects of different sizes, such as vehicles and roofs. We show the false-color image of the data set and its ground-truth map of anomalous targets in Fig. 4(b).
3) The third data set is an urban scene of Texas coast, USA, with a spatial resolution of 17.2 m. It was collected by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor [18]. The scene used in the experiment has a size of 100 $\times$ 100 pixels and 205 spectral bands, whose wavelengths range from 400 to 2500 nm.

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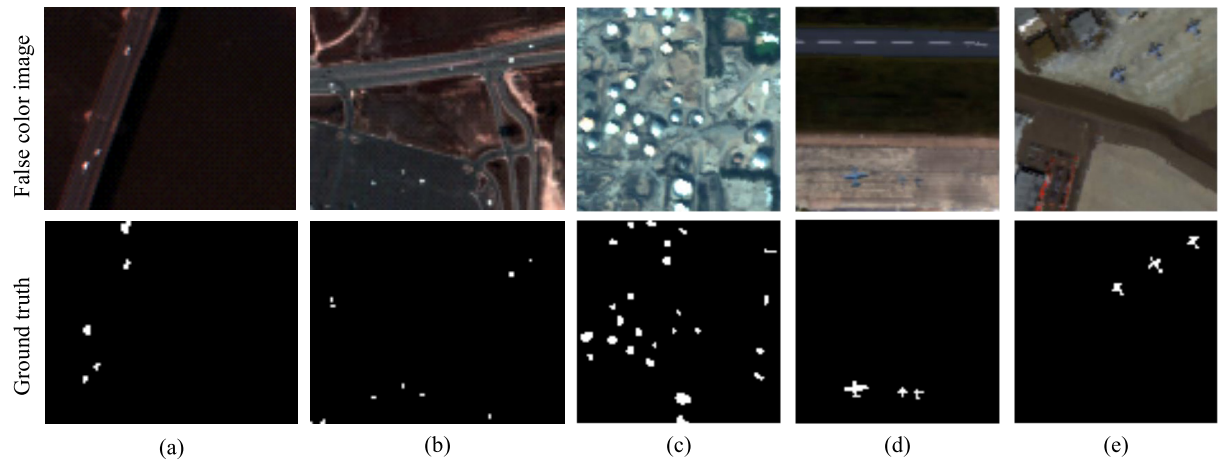


Fig. 4. Data sets used in the experiments. The first row and the second row show the false-color images and the ground-truth maps of the corresponding data sets. (a) Pavia Center. (b) HYDICE Urban. (c) Texas Coast Urban. (d) Gulfport. (e) San Diego.

We show the false-color image of the data set and its ground-truth map of anomalous targets in Fig. 4(c).

4) The fourth data set is the Gulfport data set, which was obtained by AVIRIS sensor with a spatial resolution 3.5 m and 224 spectral channels in wavelengths ranging from 400 to 2500 nm [18]. The scene used consists of 100 × 100 pixels. The bands severely affected by moisture and with the low signal-to-noise ratio (SNR) were removed, and 191 spectral bands were left. The abnormal targets are a larger plane and two small planes. We show the false-color image of the data set and its ground-truth map of anomalous targets in Fig. 4(d).
5) The fifth data set was captured by the AVIRIS sensor from San Diego, CA, USA. It has a spatial resolution of 3.5 m and a spectral resolution of 10 nm. The wavelengths range from 370 to 2510 nm and are divided into 224 spectral bands. The bands of water absorption, low SNR, and poor quality were removed, leaving 189 bands. We use a scene with a size of 100 × 100 pixels in the experiment. Three airplanes are the anomalies to be detected. We show the false-color image of the data set and its ground-truth map of anomalous targets in Fig. 4(e).

B. Comparison Methods and Evaluation Indexes

1) Comparison Methods: To thoroughly evaluate the performance of the proposed method, eight state-of-the-art hyperspectral anomaly detectors were selected for comparison, namely, GRX [38], RPCA [38], CRD [33], LRSAR [7], LSMAD [53], GTVLRR [40], FEBPAD [54], and FrFT [16]. The methods, GRX and FrFT, are statistics-based methods. The CRD is a well-known benchmark based on collaborative representation. RPCA, LSMAD, and FEBPAD are the well-known state-of-the-art methods based on low-rank and sparse matrix decomposition, and LRSAR and GTVLRR are well-known state-of-the-art methods based on LRR and sparse dictionary learning. A MATLAB demo of the proposed DeCNN-AD method will be available at https://github.com/FxyPd.

2) Evaluation Indexes: We evaluated the anomaly detection performance of the proposed method in comparison with eight algorithms qualitatively and quantitatively. We perform the qualitative evaluation by visually comparing the detection maps. Furthermore, we conduct the quantitative evaluation by using four indexes. The first index is the receiver operating characteristic (ROC) curve [55], which reflects how the probability of detection (PD) would change as the FAR varies given the ground truth. More specifically, PD and FAR can be represented as

$$\mathrm{PD} = \frac{N_{\mathrm{D}}}{N_{\mathrm{T}}}, \quad \mathrm{FAR} = \frac{N_{\mathrm{F}}}{N} \tag{27}$$

where N_{D}, N_{T}, N_{F}, and N denote the number of detected target pixels given a certain threshold, the total number of true anomalous pixels, the number of false alarm pixels, and the number of pixels in the HSI, respectively. A method is considered to be of better performance if its PD is higher than other methods at a given threshold. The second evaluating index is the area under the ROC curve (AUC), which can be derived by calculating the area under the ROC curve. The method with larger AUC is considered to obtain a better detection performance [56]. The third evaluating index is the normalized background anomaly separation map, which is depicted by a box plot. Before drawing the box plot, the values of all pixels in the anomaly maps of each algorithm are first normalized to the range of 0–1. After the normalization, the pixels of background and anomaly are extracted using the ground-truth map and depicted using the box plot. The algorithm with a larger gap between the background box and the anomaly box is considered to outperform other methods. Finally, the detecting time is also incorporated to help evaluate the computational efficiency of different algorithms.

C. Experimental Settings

Here, we discuss the parameters' setting in the proposed method and the compared methods. The parameters of the proposed anomaly detection method are K, β, and λ. The optimal settings of parameters (K, β, λ) in the proposed

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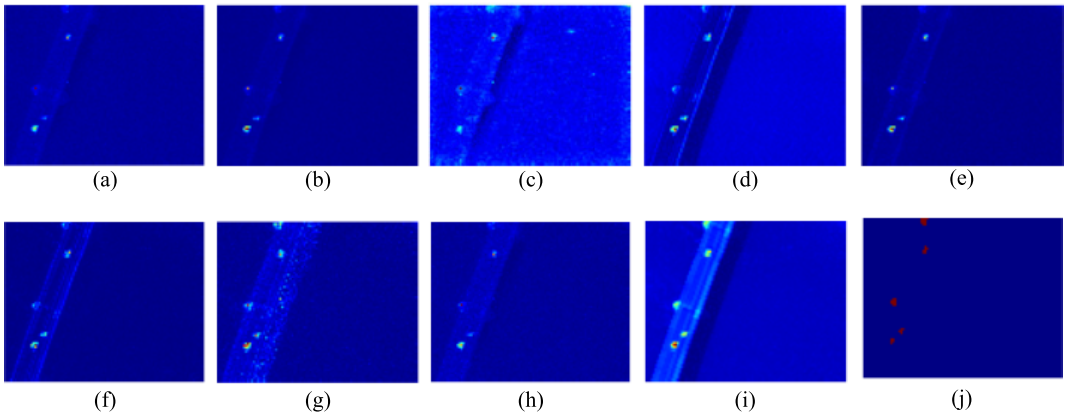


Fig. 5. Color anomaly maps obtained by different methods on the Pavia Center data set. (a) GRX. (b) RPCA. (c) CRD. (d) LRSAR. (e) LSMAD. (f) GTVLRR. (g) FEBPAD. (h) FrFT. (i) DeCNN-AD. (j) Ground truth.

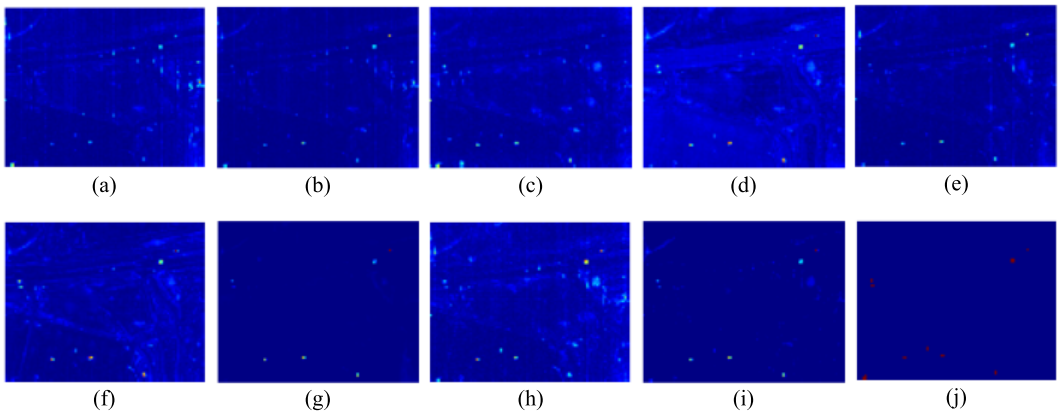


Fig. 6. Color anomaly maps obtained by different methods on the HYDICE Urban data set. (a) GRX. (b) RPCA. (c) CRD. (d) LRSAR. (e) LSMAD. (f) GTVLRR. (g) FEBPAD. (h) FrFT. (i) DeCNN-AD. (j) Ground truth.

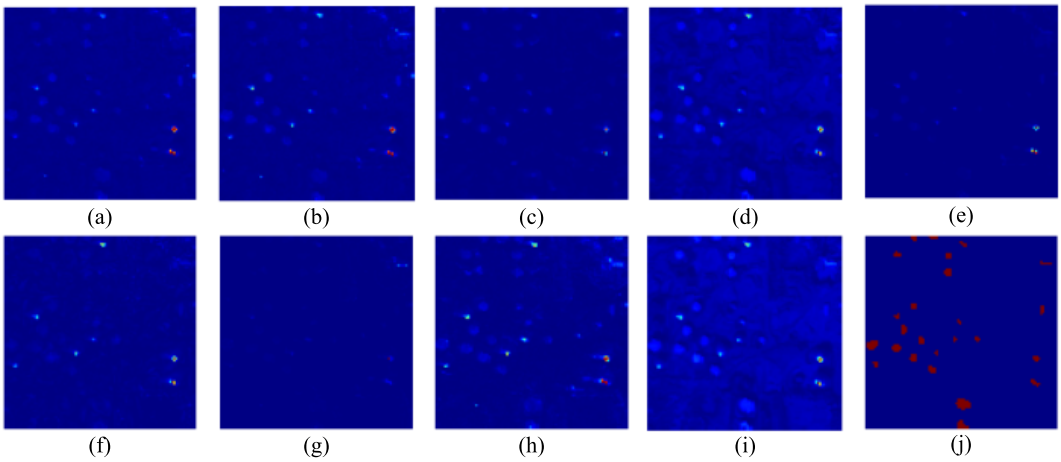


Fig. 7. Color anomaly maps obtained by different methods on the Texas Coast Urban data set. (a) GRX. (b) RPCA. (c) CRD. (d) LRSAR. (e) LSMAD. (f) GTVLRR. (g) FEBPAD. (h) FrFT. (i) DeCNN-AD. (j) Ground truth.

methods are (3, 0.001, 0.001), (4, 0.01, 0.2), (3, 0.0001, 0.0001), (6, 0.1, 0.3), and (6, 0.0001, 0.0001) for Pavia Center, HYDICE Urban, Texas Coast Urban, Gulfport, and San Diego, respectively. The regularization parameters in RPCA were all set to $\lambda = 0.01$. For the CRD method, the dual windows were set to (15, 7), (15, 11), (15, 9), (15, 13), and (17, 15) for Pavia Center, HYDICE Urban, Texas Coast Urban, Gulfport, and San Diego, respectively. The tradeoff parameters β and λ in LRSAR were set to 0.1 as recommended in [7] except that they were set to 0.001 on Texas Coast Urban data. The hyperparameters in GTVLRR were set to $\lambda = 0.5$, $\beta = 0.2$, and $\gamma = 0.05$ according to the suggestion in [40]. The settings of fractional transform order p in FEBPAD were tuned to 0.7, 0.6, 0.8, 0.6, and 0.7 to obtain the optimal detecting results for Pavia Center, HYDICE Urban, Texas Coast Urban, Gulfport, and San Diego, respectively.

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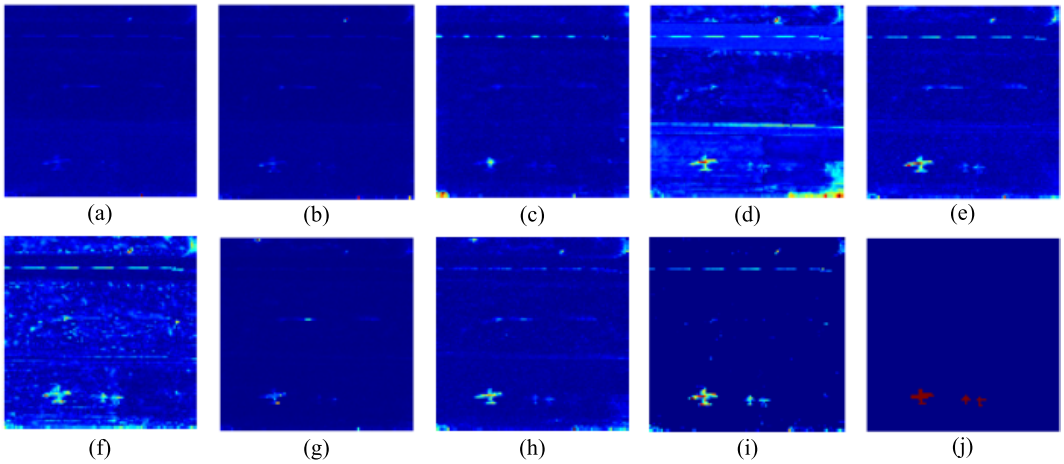


Fig. 8. Color anomaly maps obtained by different methods on the Gulfport data set. (a) GRX. (b) RPCA. (c) CRD. (d) LRSAR. (e) LSMAD. (f) GTVLRR. (g) FEBPAD. (h) FrFT. (i) DeCNN-AD. (j) Ground truth.

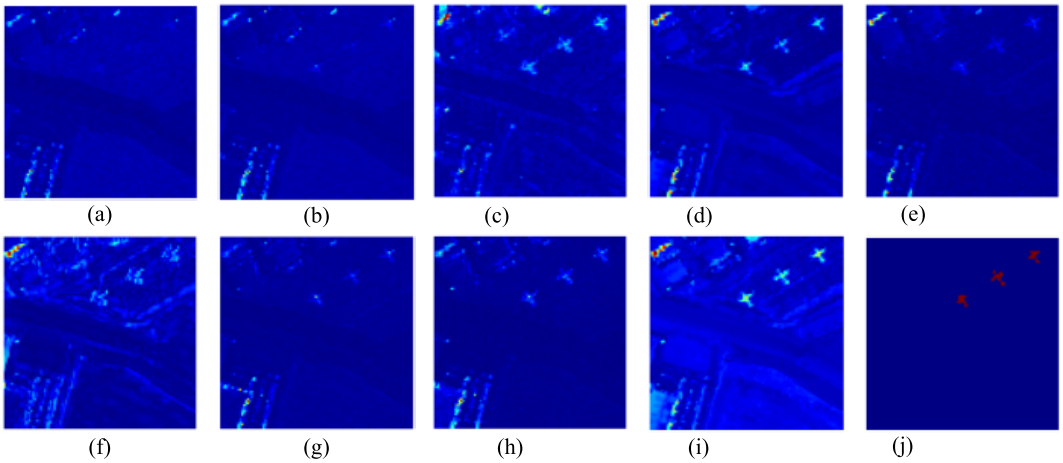


Fig. 9. Color anomaly maps obtained by different methods on the San Diego data set. (a) GRX. (b) RPCA. (c) CRD. (d) LRSAR. (e) LSMAD. (f) GTVLRR. (g) FEBPAD. (h) FrFT. (i) DeCNN-AD. (j) Ground truth.

D. Denoising Performance

The anomaly maps obtained from the different anomaly detection methods on the five experimental data sets are shown in Figs. 5–9. For the Pavia Center data, as shown in Fig. 5, although almost all the methods can detect some anomalies, the proposed DeCNN-AD method can detect the anomalies completely and obviously. For the HYDICE Urban data set, as shown in Fig. 6, LRASRR, GTVLRR, FEBPAD, FrFT, and the proposed DeCNN-AD method achieve the comparable visual results, whereas the performance of other methods is worse. For the Texas Coast Urban data set, as shown in Fig. 7, the anomalies in anomaly maps of LRASR and the proposed method can be visually recognized easier than that of the other methods. For the Gulfport data set, as shown in Fig. 8, GRX, RPCA, CRD, and FEBPAD failed to detect the complete shape of the airplanes, neither the larger planes nor two smaller airplanes. LRASR, LSMAD, GTVLRR, and FrFT can detect the larger planes with better shapes, but they cannot detect the two smaller planes effectively. Our DeCNN-AD method can detect both larger and smaller air planes more clearly and completely. For the San Diego data set, as shown in Fig 9, GRX and RPCA failed to detect the airplanes. CRD, LRSAR, and FrFT can detect the airplanes without obtaining the complete shapes. The proposed method can detect airplanes more accurately and clearly. In terms of the anomaly maps, it can be seen that the proposed method shows better detection performance than other methods.

We present ROC curves of detectors on all test images in Fig. 10. A detector is considered to perform better if its ROC curve is very close to the left upper corner of the coordinate, meaning achieving a higher detection rate and a lower FAR simultaneously. It can be observed from Fig. 10 that the proposed DeCNN-AD method outperforms other methods. From Fig. 10(a), we can see that the proposed method is superior to other methods since its ROC curve is above the ROC curves of other methods at first and then reach the top of the coordinate. In Fig. 10(b), although the detection probability of the proposed method is not always the best because it is not a little worse than CRD, FEBPAD, and GTVLRR at some range of false alarm values, it has the best performance in overall. For Fig. 10(c) and (d), the proposed method has much better ROC curves than the other methods at almost all the false alarm values. In Fig. 10(e), the ROC curve of the proposed DeCNN-AD method is the best one at most of

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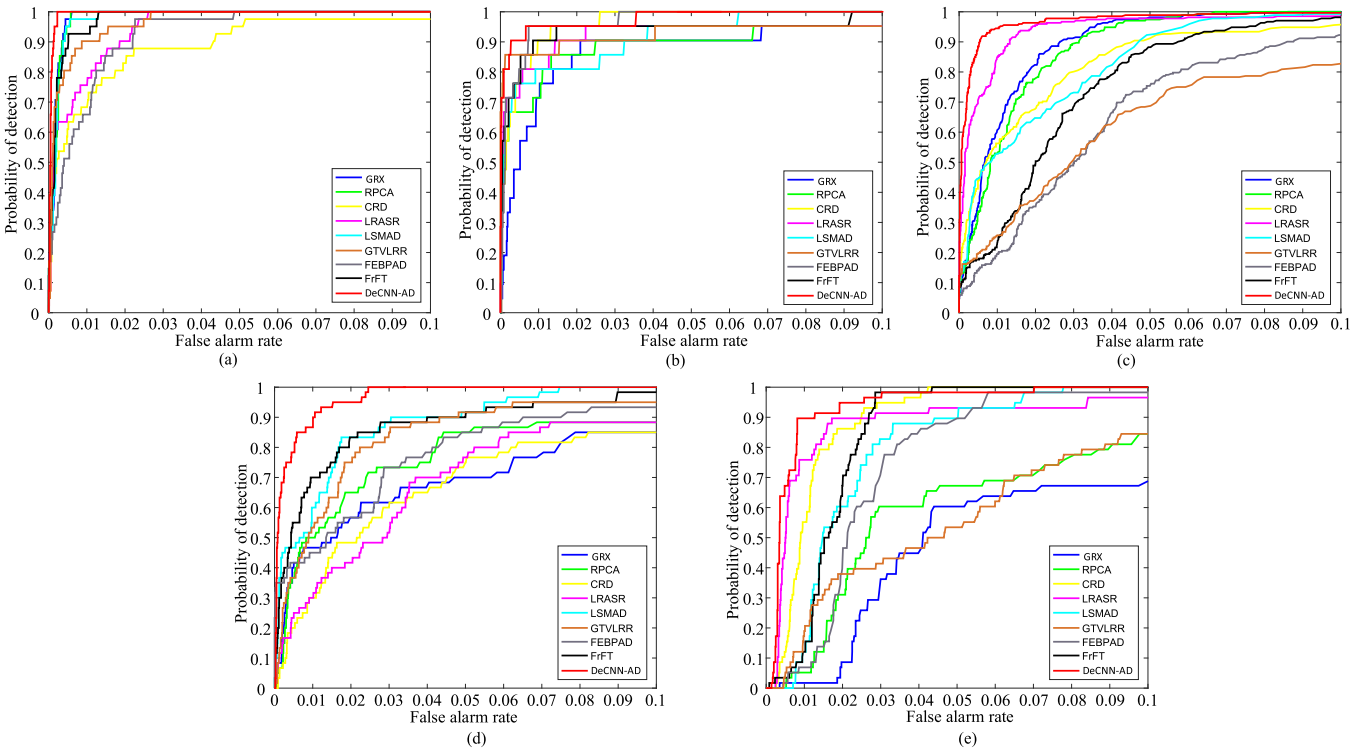


Fig. 10. ROC curves obtained by different methods on five experimental data sets. (a) Pavia Center. (b) HYDICE Urban. (c) Texas Coast Urban. (d) Gulfport. (e) San Diego.

TABLE I

AUC SCORES OF DIFFERENT DETECTORS ON FIVE EXPERIMENTAL DATA SETS

Datasets	GRX	RPCA	CRD	LRASR	LSMAD	GTVLRR	FEBPAD	FrFT	DeCNN-AD
Pavia Center	0.9984	0.9982	0.9862	0.9913	0.9979	0.9965	0.9958	0.9977	**0.9994**
HYDICE Urban	0.9857	0.9842	0.9956	0.9918	0.9903	0.9919	0.9959	0.9935	**0.9976**
Texas Coast Urban	0.9887	0.9871	0.9771	0.9920	0.9818	0.9364	0.9714	0.9740	**0.9959**
Glufport	0.9526	0.9627	0.9559	0.9521	0.9870	0.9820	0.9723	0.9852	**0.9966**
San Diego	0.8885	0.9165	0.9880	0.9886	0.9779	0.9440	0.9730	0.9827	**0.9932**

the false alarm values except for the range between 0.03 and 0.06 of false alarm where the ROC curves of CRD and FrFT are the best. In general, the performance of the proposed method in terms of the ROC curve is better than the other algorithms.

Compared with the ROC curve, the AUC score is more straightforward to compare the performance of different detectors. To further compare the performance of the proposed DeCNN-AD method and the comparison methods, we calculated the AUC scores of all the methods on five experimental data sets. The detailed scores are presented in Table I, and the best AUC scores are highlighted in bold. It can be observed from Table I that the proposed DeCNN-AD method achieves the best results on all the experimental data sets. For the Pavia Center data set and the HYDICE Urban data set in which the sizes of anomalies are relatively small, all the methods achieve good detecting performance. The AUCs of the proposed DeCNN-AD method are 0.9994 and 0.9976, respectively. For the Texas Coast Urban, the Gulfport, and the San Diego data sets in which the anomalies are planar, the AUCs of the proposed DeCNN-AD method are 0.9959, 0.9966, and 0.9932, respectively, whereas the other methods achieve relatively lower AUCs in comparison with the proposed method.

Box plots are used to facilitate a more in-depth evaluation of the separability between the background and the anomalies for all the methods on five data sets. The detailed results are shown in Fig. 11, where we use the red boxes to denote the anomaly score distributions of anomaly pixels and green boxes to present the distributions of the background pixels. The relative position of the red and green box represents the separability between anomaly pixels and background pixels. More specifically, the method with a larger distance between the background box and anomaly box is considered to achieve a more superior detection performance. From Fig. 11, we can observe that the proposed DeCNN-AD method obtains better background and anomaly separation than GRX, RPCA, CRD, LRASR, and LSMAD on all the five experimental data sets. FrFT and GTVLRR achieve better separation than the proposed method on the HYDICE Urban data set, as shown in Fig. 11(b), and FEBPAD achieves a better separation than the proposed DeCNN-AD method on the Texas Coast Urban

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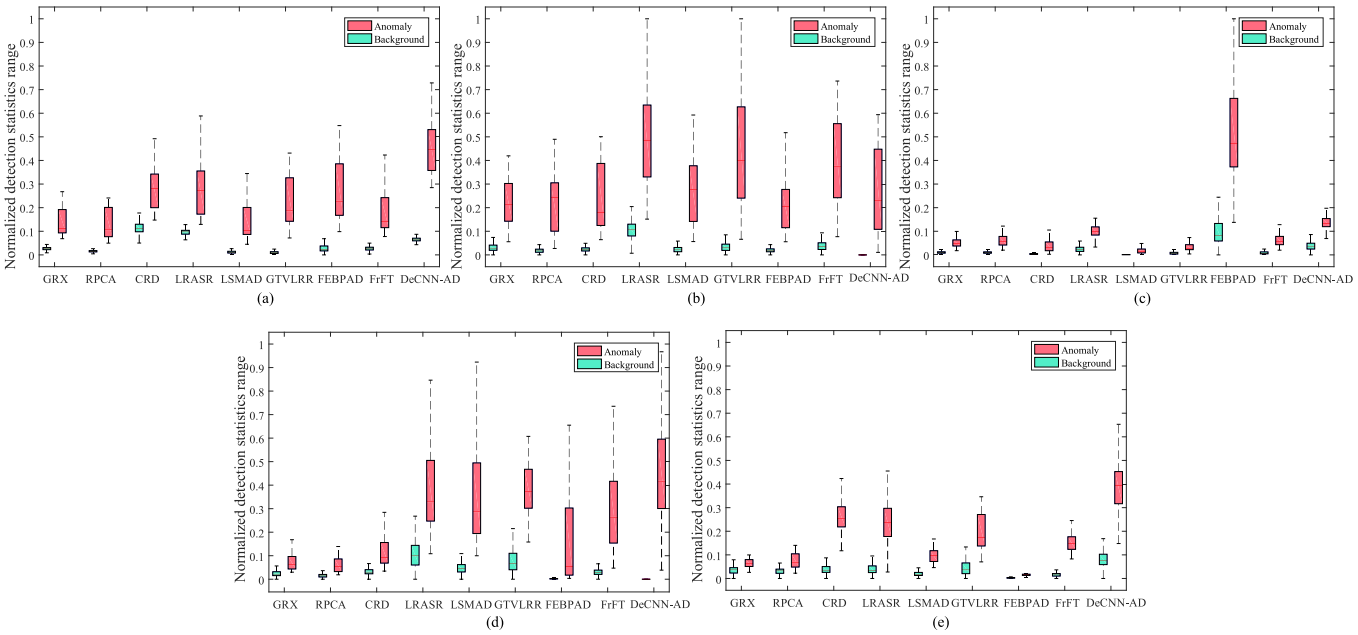


Fig. 11. Box plots obtained by different methods on five experimental data sets. (a) Pavia Center. (b) HYDICE Urban. (c) Texas Coast Urban. (d) Gulfport. (e) San Diego.

TABLE II
RUNNING TIME OF DIFFERENT DETECTORS ON FIVE EXPERIMENTAL DATA SETS (IN SECONDS)

Datasets	GRX	RPCA	CRD	LRASR	LSMAD	GTVLRR	FEBPAD	FrFT	DeCNN-AD
Pavia Center	0.07	3.62	35.14	40.66	6.09	216.48	2.01	8.22	22.85
HYDICE Urban	0.10	3.60	59.83	49.15	9.84	302.97	2.93	11.03	28.88
Texas Coast Urban	0.13	9.15	84.67	42.82	15.11	321.46	3.93	15.93	17.55
Gulfport	0.12	5.16	12.62	53.49	13.88	386.19	4.14	14.78	44.27
San Diego	0.19	5.41	10.97	53.81	16.43	346.24	4.01	14.17	43.13

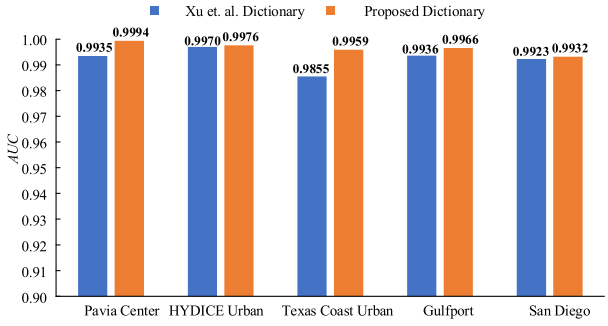


Fig. 12. AUC of the proposed DeCNN-AD method by using different dictionary construction methods on five data sets.

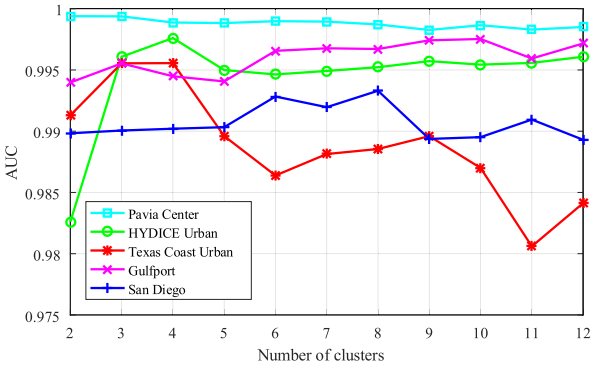


Fig. 13. AUC of the proposed DeCNN-AD method with the change of K.

data set, as shown in Fig. 11(c). However, the results of these methods show worse separations of the background and anomalies on the other four data sets. Overall, we can conclude that the proposed DeCNN-AD method can achieve a better separation between background and anomaly than other comparison algorithms.

E. Parameter Analysis

In terms of the ROC, AUC, and the separation between normalized background and anomaly, the abovementioned observations suggest that the proposed DeCNN-AD method is effective in anomaly detection in HSI and performs better than the other methods on five experimental data sets in overall.

Finally, the running times of the proposed DeCNN-AD method and eight comparison methods are compared in MATLAB (R2016a) on Windows 10 with an Intel Core i7-7700HQ 2.80-GHz processor and 24-GB RAM. We recorded the running time of all the detection methods on five experimental data sets. The detailed results are presented in Table II, where we can observe that GRX took the shortest time to perform anomaly detecting. RPCA and FEBPAD are also very efficient in computation. Due to the

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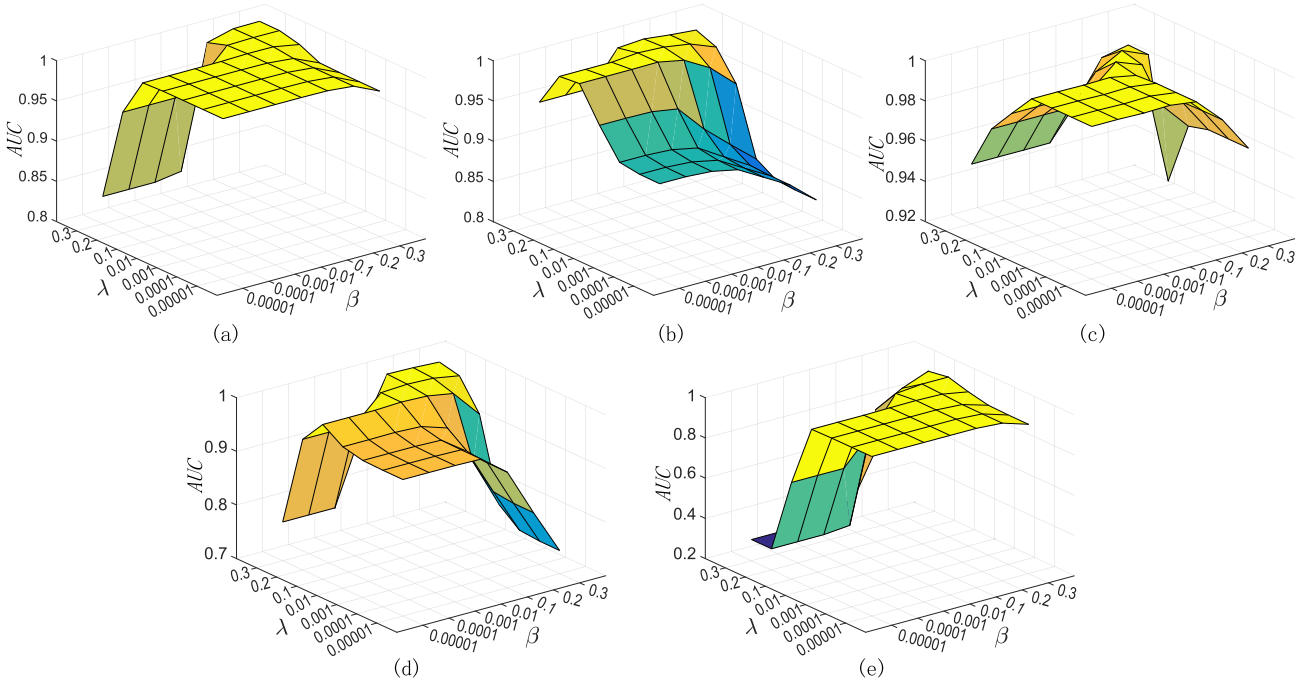


Fig. 14. AUC of the proposed DeCNN-AD method with the change of β and λ. (a) Pavia Center. (b) HYDICE Urban. (c) Texas Coast Urban. (d) Gulfport. (e) San Diego.

complex regulation of the representation coefficient, GTVLRR takes the longest time to detect anomalies. Compared with the LRR-based anomaly detection methods, i.e., LRASR and GTVLRR, our DeCNN-AD method took less time to detect anomalies on all the five data sets.

F. Performance of Background Dictionary Construction

In order to verify the effectiveness of the proposed dictionary construction method, we compared the detection performance of the proposed method in five data sets by using a dictionary construction method proposed in [7] and our proposed dictionary construction method. We derived the AUC of the detection maps in the five data sets, and the results are shown in Fig. 12, where we can see that the proposed dictionary construction method could help improve the detection performance in five data sets to varying degrees. The proposed dictionary construction method eliminates the potential abnormal pixels using inner iterations within each cluster, which helps lower the probability of dictionary being contaminated by anomalous pixels. Furthermore, by adding the pixels with the smallest Mahalanobis distance to the center of each cluster, the constructed background dictionary becomes more representative, which also helps improve the detection performance.

The parameters needed to be tuned in the proposed DeCNN-AD method is the cluster number K and the tradeoff parameters β and λ. The cluster number K is used to cluster the HSI into K groups to construct the background dictionary. The detection performance (in terms of AUC) of the proposed DeCNN-AD method with the change of K in five data sets are presented in Fig. 13. We can see from the figure that the detection performance for all the data sets remain in a high value and show little fluctuation except for the San Diego data set, whose detection performance is good at less clusters and descends with fluctuations as the cluster number increases. Thus, K with the optimal results was selected in the experiments.

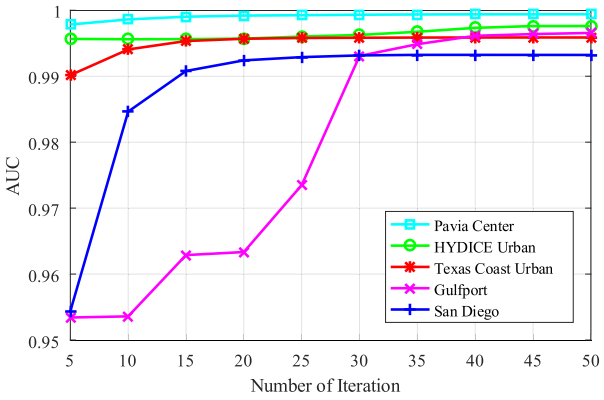


Fig. 15. AUC of the proposed DeCNN-AD method with the change of iteration numbers.

The parameters β and λ tradeoff the implicit regulation on $\mathbf{S}$ and the group sparsity regulation on $\mathbf{E}$, respectively. They play important role in the anomaly detection. To evaluate the sensitivity of the proposed DeCNN-AD method to the parameters, the detection performance with the change of β and λ in five data sets is presented in Fig. 14, where we can see that, for Pavia data, Texast Coast data, and San Diego data, the detection performance is good with no obvious change when $\beta < 0.1$ and $\lambda < 0.01$. For the HYDICE urban data and the GulfPort data, the detection performance is good when $\beta \geq 0.01$ and $\lambda \geq 0.1$. The optimal values of β and λ may

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vary for different data. However, a good detecting results can also be obtained within the range of β and λ. Thus, we suggest choosing β and λ from the parameter range carefully to obtain the desired detecting results.

In order to assess the detecting stability with the change of iteration numbers for the proposed DeCNN-AD method, the change of AUC with the number of iterations on five data sets are shown in Fig. 15. It can be seen in the figure that the AUCs of all the data set reach the optimal results with no obvious change after 40 iterations; thus, a maximum iteration number larger than 40 can ensure the desired detecting results. Taking the running time and the detecting performance into consideration, the maximum iteration number is fixed to 50 in our experiments.

V. Conclusion

In this article, we proposed a novel anomaly detection method plugged with a CNN regularization. To avoid cumbersomely handcrafting a regularizer, we use the plug-and-play framework, which enables us to use a flexible denoiser as the prior for the dictionary representation coefficients. FFDNet, a state-of-the-art CNN-based denoiser, is plugged to efficiently constrain the representation coefficients. We remark that our detection framework also has the potential to be plugged with other kinds of denoisers conceived for image denoising. What is more, to obtain the background dictionary with better purity, we proposed a modified dictionary construction method. We performed extensive experiments with five data sets in a comparison with eight state-of-the-art anomaly detection methods. The detection performances were evaluated qualitatively and quantitatively in terms of visibility, ROC curve, AUC, and the normalized background anomaly separation map. Experimental results suggest that the proposed method is effective in anomaly detection and can achieve better detecting results than the compared methods. Considering the complex characteristics of the representation coefficients in the LRR-based methods and the strong learning ability of deep learning, elaborating a new deep prior rather than using existing denoising prior to fully exploit the information of the representation coefficients, may help to promote the performance of the anomaly detection.

References

[1] J. M. Bioucas-Dias *et al.*, "Hyperspectral unmixing overview: Geometrical, statistical, and sparse regression-based approaches," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 354–379, Apr. 2012.

[2] X. Lu, Y. Yuan, and X. Zheng, "Joint dictionary learning for multispectral change detection," *IEEE Trans. Cybern.*, vol. 47, no. 4, pp. 884–897, Apr. 2017.

[3] L. Zhuang, C. Lin, M. A. T. Figueiredo, and J. M. Bioucas-Dias, "Regularization parameter selection in minimum volume hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9858–9877, Dec. 2019.

[4] M. T. Eismann, J. Meola, and R. C. Hardie, "Hyperspectral change detection in the presenceof diurnal and seasonal variations," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 1, pp. 237–249, Jan. 2008.

[5] Y. Zhang, B. Du, L. Zhang, and T. Liu, "Joint sparse representation and multitask learning for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 894–906, Feb. 2017.

[6] D. W. J. Stein, S. G. Beaven, L. E. Hoff, E. M. Winter, A. P. Schaum, and A. D. Stocker, "Anomaly detection from hyperspectral imagery," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 58–69, Jan. 2002.

[7] Y. Xu, Z. Wu, J. Li, A. Plaza, and Z. Wei, "Anomaly detection in hyperspectral images based on low-rank and sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 1990–2000, Apr. 2016.

[8] H. Su, Z. Wu, A.-X. Zhu, and Q. Du, "Low rank and collaborative representation for hyperspectral anomaly detection via robust dictionary construction," *ISPRS J. Photogramm. Remote Sens.*, vol. 169, pp. 195–211, Nov. 2020.

[9] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Oct. 1990.

[10] Q. Guo, B. Zhang, Q. Ran, L. Gao, J. Li, and A. Plaza, "Weighted-RXD and linear filter-based RXD: Improving background statistics estimation for anomaly detection in hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2351–2366, Jun. 2014.

[11] H. Kwon and N. M. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005.

[12] M. J. Carlotto, "A cluster-based approach for detecting man-made objects and changes in imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 374–387, Feb. 2005.

[13] L. Gao, Q. Guo, A. Plaza, J. Li, and B. Zhang, "Probabilistic anomaly detector for remotely sensed hyperspectral data," *J. Appl. Remote Sens.*, vol. 8, no. 1, Nov. 2014, Art. no. 083538.

[14] A. Schaum, "Joint subspace detection of hyperspectral targets," in *Proc. IEEE Aerosp. Conf.*, Mar. 2004, pp. 1818–1824.

[15] B. Du and L. Zhang, "A discriminative metric learning based anomaly detection method," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 11, pp. 6844–6857, Nov. 2014.

[16] R. Tao, X. Zhao, W. Li, H.-C. Li, and Q. Du, "Hyperspectral anomaly detection by fractional Fourier entropy," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4920–4929, Dec. 2019.

[17] B. Tu, X. Yang, C. Zhou, D. He, and A. Plaza, "Hyperspectral anomaly detection using dual window density," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8503–8517, Dec. 2020.

[18] X. Kang, X. Zhang, S. Li, K. Li, J. Li, and J. A. Benediktsson, "Hyperspectral anomaly detection with attribute and edge-preserving filters," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5600–5611, Oct. 2017.

[19] S. Li, K. Zhang, Q. Hao, P. Duan, and X. Kang, "Hyperspectral anomaly detection with multiscale attribute and edge-preserving filters," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 10, pp. 1605–1609, Oct. 2018.

[20] S. Li, K. Zhang, P. Duan, and X. Kang, "Hyperspectral anomaly detection with kernel isolation forest," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 319–329, Jan. 2020.

[21] R. Wang, F. Nie, Z. Wang, F. He, and X. Li, "Multiple features and isolation forest-based fast anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 9, pp. 6664–6676, Sep. 2020.

[22] J. Lei, S. Fang, W. Xie, Y. Li, and C.-I. Chang, "Discriminative reconstruction for hyperspectral anomaly detection with spectral learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7406–7417, Oct. 2020.

[23] S. Chang, B. Du, and L. Zhang, "A subspace selection-based discriminative forest method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 4033–4046, Jun. 2020.

[24] S. Wang, X. Wang, Y. Zhong, and L. Zhang, "Hyperspectral anomaly detection via locally enhanced low-rank prior," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 6995–7009, Oct. 2020.

[25] Y. Gao, T. Cheng, and B. Wang, "Nonlinear anomaly detection based on spectral-spatial composite kernel for hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, early access, May 25, 2020, doi: 10.1109/LGRS.2020.2994629.

[26] L. Li, W. Li, Y. Qu, C. Zhao, R. Tao, and Q. Du, "Prior-based tensor approximation for anomaly detection in hyperspectral imagery," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Dec. 9, 2020, doi: 10.1109/TNNLS.2020.3038659.

[27] Z. Li, Q. Ling, Z. Lin, and J. Wu, "Segmentation-based weighting strategy for hyperspectral anomaly detection," *IEEE Geosci. Remote Sens. Lett.*, early access, Jul. 22, 2020, doi: 10.1109/LGRS.2020.3009147.

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[28] S. Song, H. Zhou, Y. Yang, and J. Song, "Hyperspectral anomaly detection via convolutional neural network and low rank with density-based clustering," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 9, pp. 3637–3649, Sep. 2019.
[29] W. Xie *et al.*, "Unsupervised spectral mapping and feature selection for hyperspectral anomaly detection," *Neural Netw.*, vol. 132, pp. 144–154, Dec. 2020.
[30] W. Xie, B. Liu, Y. Li, J. Lei, and Q. Du, "Autoencoder and adversarial-learning-based semisupervised background estimation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5416–5427, Aug. 2020.
[31] W. Li, G. Wu, and Q. Du, "Transferred deep learning for anomaly detection in hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 597–601, May 2017.
[32] L. Zhang and B. Cheng, "Transferred CNN based on tensor for hyperspectral anomaly detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 12, pp. 2115–2119, Dec. 2020.
[33] W. Li and Q. Du, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.
[34] M. Imani, "Anomaly detection using morphology-based collaborative representation in hyperspectral imagery," *Eur. J. Remote Sens.*, vol. 51, no. 1, pp. 457–471, Jan. 2018.
[35] H. Su, Z. Wu, Q. Du, and P. Du, "Hyperspectral anomaly detection using collaborative representation with outlier removal," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 12, pp. 5029–5038, Dec. 2018.
[36] N. Ma, Y. Peng, and S. Wang, "A fast recursive collaboration representation anomaly detector for hyperspectral image," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 4, pp. 588–592, Apr. 2019.
[37] R. Wang, H. Hu, F. He, F. Nie, S. Cai, and Z. Ming, "Self-weighted collaborative representation for hyperspectral anomaly detection," *Signal Process.*, vol. 177, Dec. 2020, Art. no. 107718.
[38] S.-Y. Chen, S. Yang, K. Kalpakis, and C.-I. Chang, "Low-rank decomposition-based anomaly detection," *Proc. SPIE, Algorithms Technol. Multispectral, Hyperspectral, Ultraspectral Imag. XIX*, vol. 8743, May 2013, Art. no. 87430N.
[39] G. Liu, Z. Lin, S. Yan, J. Sun, Y. Yu, and Y. Ma, "Robust recovery of subspace structures by low-rank representation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 1, pp. 171–184, Jan. 2013.
[40] T. Cheng and B. Wang, "Graph and total variation regularized low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 391–406, Jan. 2020.
[41] K. Dabov, A. Foi, V. Katkovnik, and K. Egiazarian, "Image denoising by sparse 3-D transform-domain collaborative filtering," *IEEE Trans. Image Process.*, vol. 16, no. 8, pp. 2080–2095, Aug. 2007.
[42] S. Gu, L. Zhang, W. Zuo, and X. Feng, "Weighted nuclear norm minimization with application to image denoising," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 2862–2869.
[43] K. Zhang, W. Zuo, and L. Zhang, "FFDNet: Toward a fast and flexible solution for CNN-based image denoising," *IEEE Trans. Image Process.*, vol. 27, no. 9, pp. 4608–4622, Sep. 2018.
[44] K. Zhang, W. Zuo, Y. Chen, D. Meng, and L. Zhang, "Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising," *IEEE Trans. Image Process.*, vol. 26, no. 7, pp. 3142–3155, Jul. 2017.
[45] A. M. Teodoro, J. M. Bioucas-Dias, and M. A. T. Figueiredo, "Image restoration and reconstruction using variable splitting and class-adapted image priors," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2016, pp. 3518–3522.
[46] L. Zhuang and J. M. Bioucas-Dias, "Fast hyperspectral image denoising and inpainting based on low-rank and sparse representations," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 3, pp. 730–742, Mar. 2018.
[47] L. Zhuang and M. K. Ng, "Hyperspectral mixed noise removal by ℓ_1-norm-based subspace representation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1143–1157, 2020.
[48] R. Dian, S. Li, and X. Kang, "Regularizing hyperspectral and multispectral image fusion by CNN denoiser," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Apr. 17, 2020, doi: 10.1109/TNNLS.2020.2980398.
[49] S. V. Venkatakrishnan, C. A. Bouman, and B. Wohlberg, "Plug-and-play priors for model based reconstruction," in *Proc. IEEE Global Conf. Signal Inf. Process.*, Dec. 2013, pp. 945–948.
[50] M. Maggioni, V. Katkovnik, K. Egiazarian, and A. Foi, "Nonlocal transform-domain filter for volumetric data denoising and reconstruction," *IEEE Trans. Image Process.*, vol. 22, no. 1, pp. 119–133, Jan. 2013.
[51] A. Buades, B. Coll, and J.-M. Morel, "A non-local algorithm for image denoising," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2005, pp. 60–65.
[52] J. M. Bioucas-Dias and J. M. P. Nascimento, "Hyperspectral subspace identification," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 8, pp. 2435–2445, Aug. 2008.
[53] Y. Zhang, B. Du, L. Zhang, and S. Wang, "A low-rank and sparse matrix decomposition-based mahalanobis distance method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1376–1389, Mar. 2016.
[54] Y. Ma, G. Fan, Q. Jin, J. Huang, X. Mei, and J. Ma, "Hyperspectral anomaly detection via integration of feature extraction and background purification," *IEEE Geosci. Remote Sens. Lett.*, early access, Jun. 12, 2020, doi: 10.1109/LGRS.2020.2998809.
[55] J. Kerekes, "Receiver operating characteristic curve confidence intervals and regions," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 2, pp. 251–255, Apr. 2008.
[56] T. Fawcett, "An introduction to ROC analysis," *Pattern Recognit. Lett.*, vol. 27, no. 8, pp. 861–874, Jun. 2006.



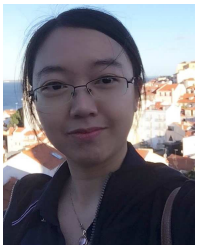
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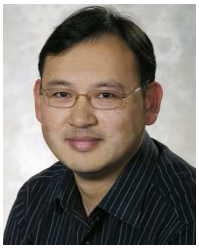
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